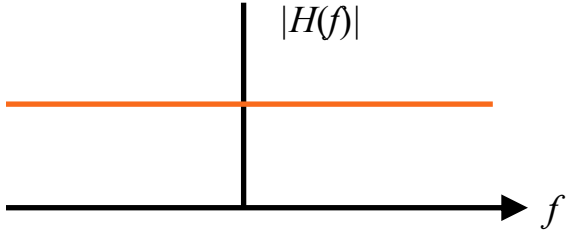
Lecture #12

EE 471C / EE 381K-17 Wireless Communication Lab

From last lecture

- Frequency flat wireless channel

Single-path channel $h_c(t) = \alpha\delta(t - \tau_d)$



BB equivalent $\longrightarrow$ $h(t) = B\alpha e^{-j2\pi f_c \tau_d} \text{sinc}(B(t - \tau_d))$

- How to deal with impairments in a frequency flat channel
  - Frame synchronization
  - Channel estimation and equalization
  - Detection
  - Frequency offset correction

Preview of today's lecture

- Frequency selective channel (going beyond frequency flat)
  - Explain different impairments caused by the frequency selective wireless communication channel
- Least-squares FIR equalizer from a channel estimate
  - Derive and compute the least-squares equalizer using an estimate of the channel
- Least minimum mean squared (LMMSE) equalizer
  - Derive and compute the LMMSE equalizer
- Single carrier frequency domain equalization
  - Motivate equalization in the frequency domain
  - Derive the SC-FDE equalizer

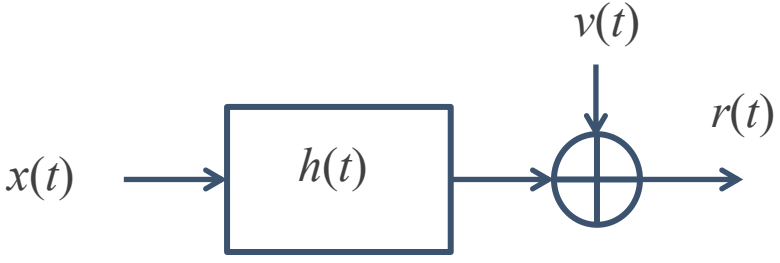
Impairments in frequency selective channels

Learning objectives:

- Explain different impairments caused by the frequency selective wireless communication channel

Frequency selective channels

- The channel between TX and RX is not just a delay and scaling
- More generally, the complex baseband equivalent channel is represented by an impulse response



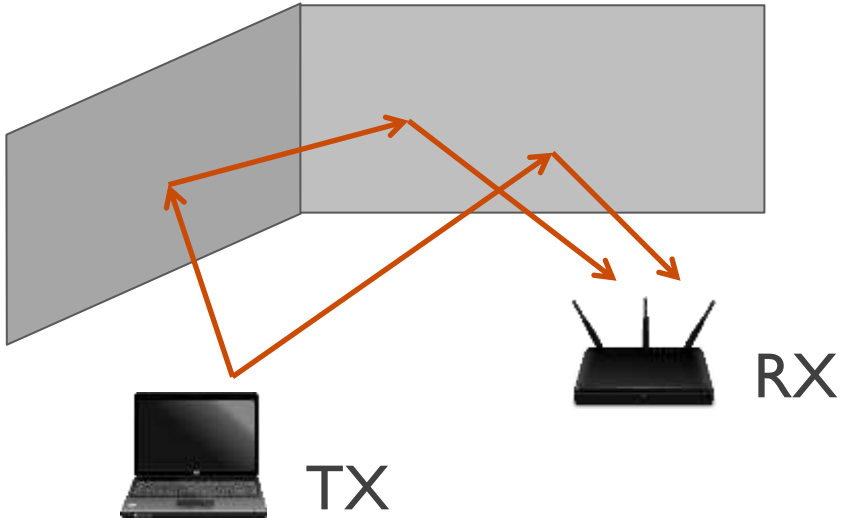
  - Called frequency selective because in general $H(f)$ is not flat (in BW of $x(t)$)
  - A reasonable assumption about $h(t)$ is that it is causal and FIR

Multipath channel

Complex gain for each path

Delay for each path

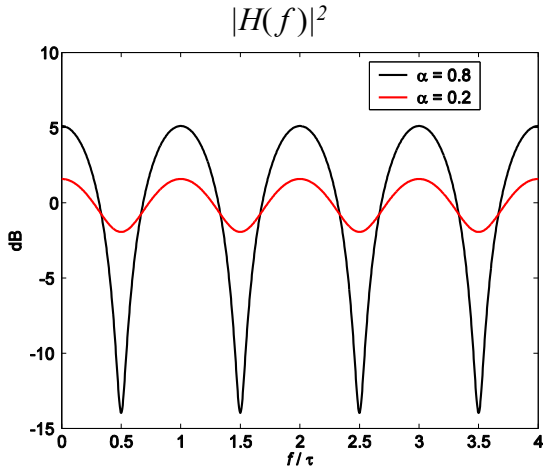
$$h(t) = \sum_k \alpha_k e^{j\phi_k} \delta(t - \tau_k)$$



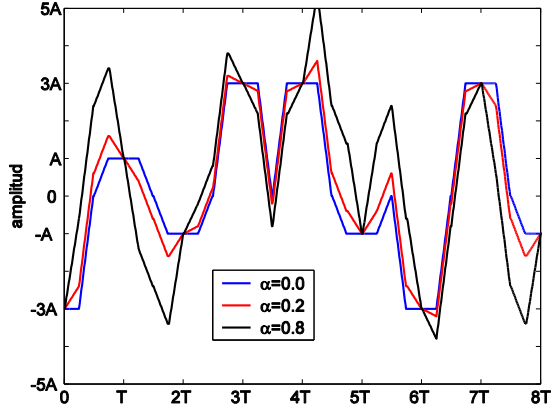
Example of multipath channel

$$h(t)=\delta(t)+\alpha\delta(t-\tau)$$

$$|H(f)|^2=1+\alpha^2+\alpha\cos(2\pi f\tau)$$



- Effect in the received signal
  - 4-PAM signal
  - $\tau = 3T/2$



Model for the received signal

- Consider the received signal after matched filtering and sampling

$$y(t) = g_{\text{rx}}(t) * h(t) * \sqrt{E_{\text{x}}} \sum_{m=-\infty}^{\infty} s[m] g_{\text{tx}}(t - mT) + g_{\text{rx}}(t) * v(t)$$

- Define the effective channel $h_{\text{eff}} = g_{\text{rx}}(t) * g_{\text{tx}}(t) * \sqrt{E_{\text{x}}} h(t)$

- Then the received signal model becomes

$$y(t) = \sum_{m=-\infty}^{\infty} s[m] h_{\text{eff}}(t - mT) + g_{\text{rx}}(t) * v(t)$$

Looks like complex pulse amplitude modulation with a new pulse shape that includes the channel

Model for the received signal

- Sampling at the symbol rate

$$y[n] = \sum_{\ell=-\infty}^{\infty} h[\ell]s[n-\ell] + v[n]$$

- Suppose the channel is causal
  - Noncausal filters are implemented with delay, propagation is causal
- Suppose the channel is FIR
  - Signal decays with distance, meaning long reflections are very weak
- The simplified system with FIR

Order of the FIR channnel, usually assumed known

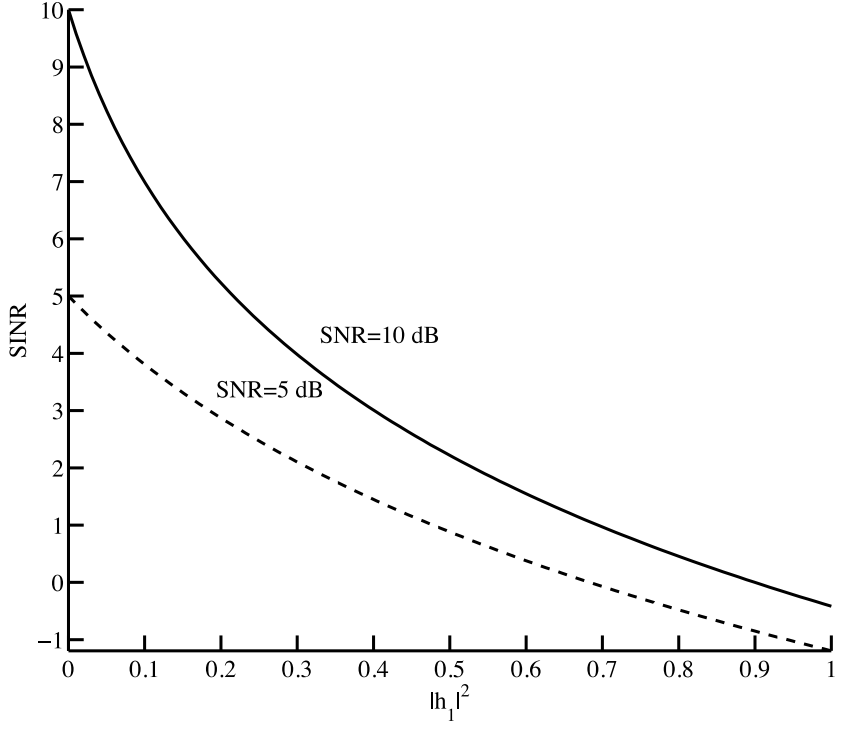
$$y[n] = \sum_{\ell=0}^{L} h[\ell]s[n-\ell] \; + \; v[n]$$

Example

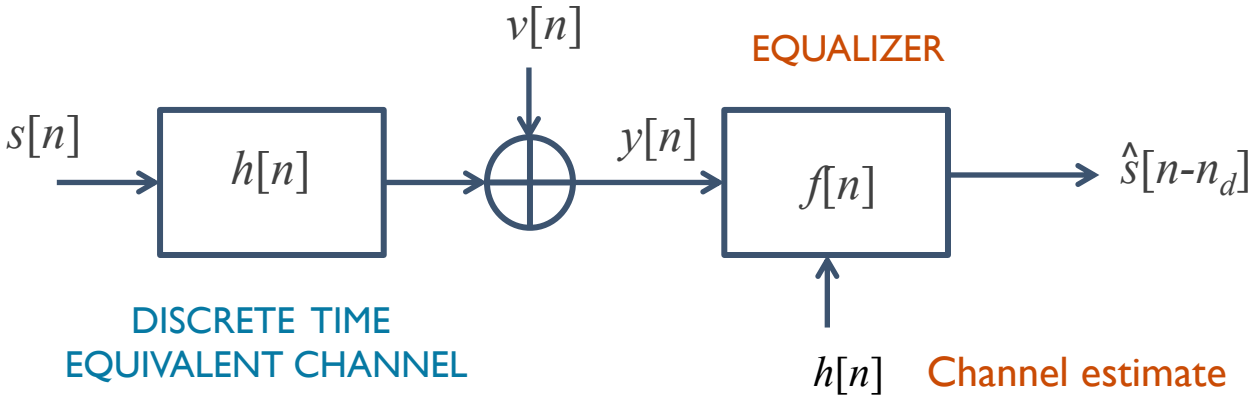
- Suppose that $h[n] = \sqrt{E_x}\delta[n] + \sqrt{E_x}h_1\delta[n-1]$
  - This is a two-tap discrete-time channel

$$y[n] = \sqrt{E_x}s[n] + \sqrt{E_x}h_1 s[n-1] + v[n]$$

$$\text{SINR} = \frac{E_x}{E_x|h_1|^2 + N_o}$$



Removing ISI using linear equalization



- An equalizer is a filter to remove or reduce ISI
  - Find the equalizer assuming the channel is known
- Other equalization solutions are possible
  - Decision feedback – combines detection and equalization
  - Maximum likelihood sequence detection – much higher complexity

Impairments in frequency selective channels

Learning objectives:

- Explain different impairments caused by the frequency selective wireless communication channel

Least squares equalizer

Learning objectives:

- Develop LS equalizers to remove the effects of intersymbol interference

Removing ISI using linear equalization

- Consider an FIR linear equalizer with coefficients $\{f[\ell]\}_{\ell=0}^{L_\mathrm{f}}$

$$\sum_{\ell=0}^{L_\mathrm{f}} f[\ell]h[n-\ell] = \delta[n-n_\mathrm{d}] \quad \text{for} \quad n = 0, 1, \ldots, L_\mathrm{f}+L$$

- Write as a linear system

$$\begin{bmatrix} h[0] & 0 & \cdots & \cdots & 0 \\ h[1] & h[0] & 0 & \cdots & \vdots \\ \vdots & \ddots & & & h[0] \\ h[\ell] & & & & h[1] \\ 0 & h[\ell] & \cdots & & \vdots \\ \vdots & & & & h[\ell] \end{bmatrix} \begin{bmatrix} f[0] \\ f[1] \\ \vdots \\ f[L_\mathrm{f}] \end{bmatrix} = \mathbf{e}_{n_\mathrm{d}} \quad \Rightarrow \quad \mathbf{H}\mathbf{f}_{n_\mathrm{d}} = \mathbf{e}_{n_\mathrm{d}}$$

Toeplitz structure

$L_f+L+1 \times L_f+1$ (tall)

Computation of the LS equalizer

- Toeplitz structure in $\mathbf{H}$ leads to efficient algorithms to solve LS
- $\mathbf{H}$ is full rank as long as at least one coefficient is nonzero
- The LS solution assuming $\mathbf{H}$ is full rank is

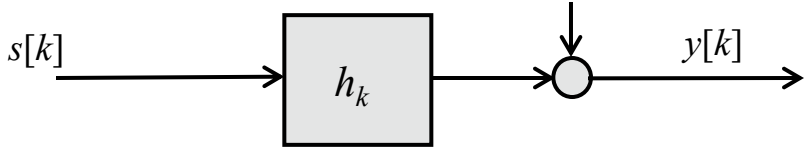
$$\mathbf{f}_{\mathrm{LS},n_\mathrm{d}} = (\mathbf{H}^*\mathbf{H})^{-1}\mathbf{H}^*\mathbf{e}_{n_\mathrm{d}}$$

  - with squared error

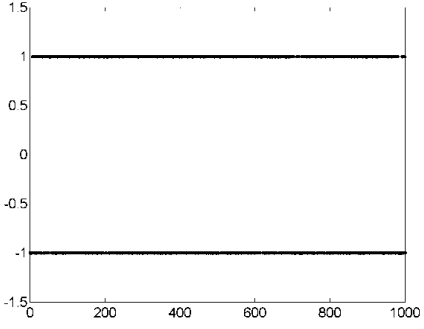
$$J[n_\mathrm{d}] = \mathbf{e}_{n_\mathrm{d}}^*(\mathbf{I} - \mathbf{H}(\mathbf{H}^*\mathbf{H})^{-1}\mathbf{H}^*)\mathbf{e}_{n_\mathrm{d}}$$

- The squared error can be further minimized by choosing n_d

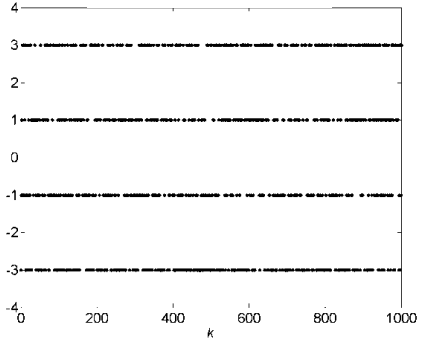
Example: ISI channel



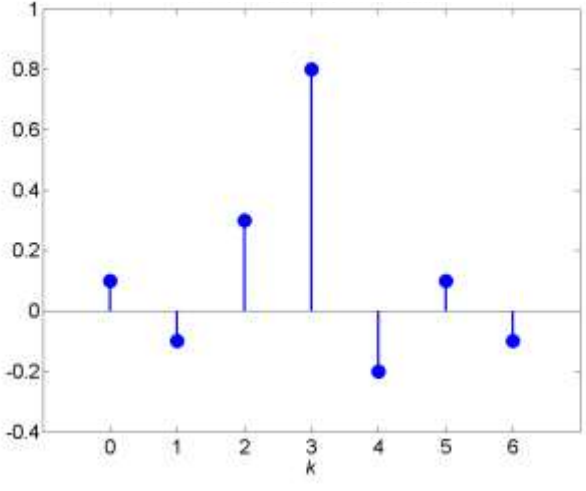
BPSK $s[k]$



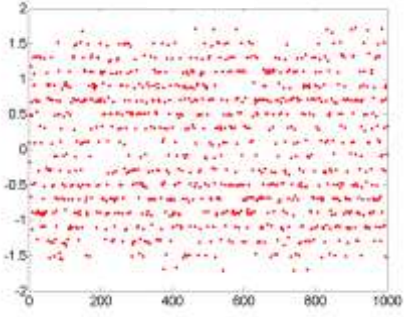
4-PAM $s[k]$



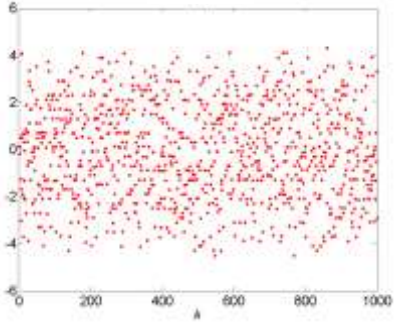
channel $h[k]$



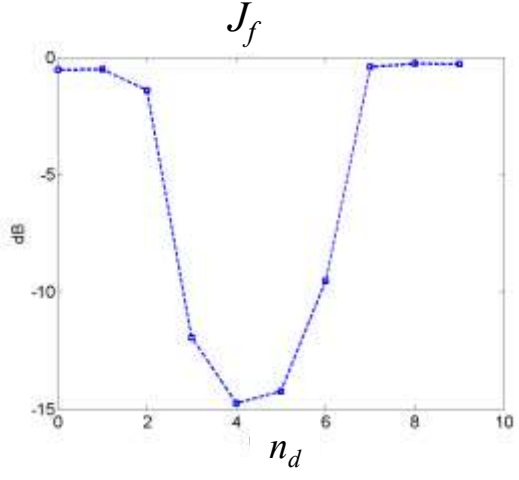
$y[k]$



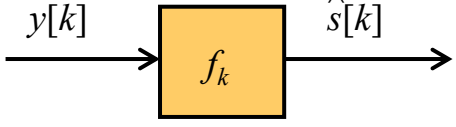
$y[k]$

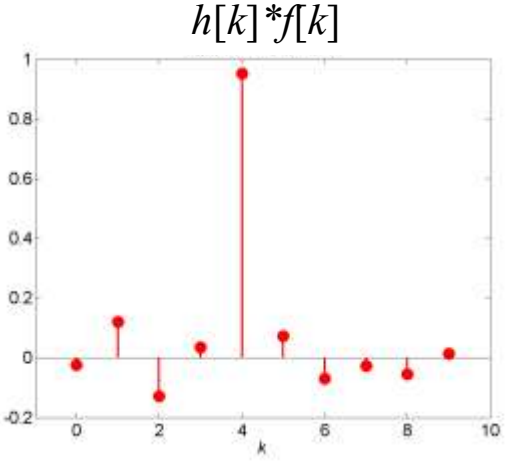


LS equalizer, order 3

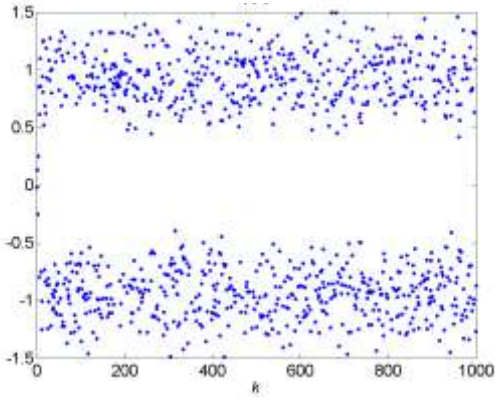


optimal delay
$n_d = 4$

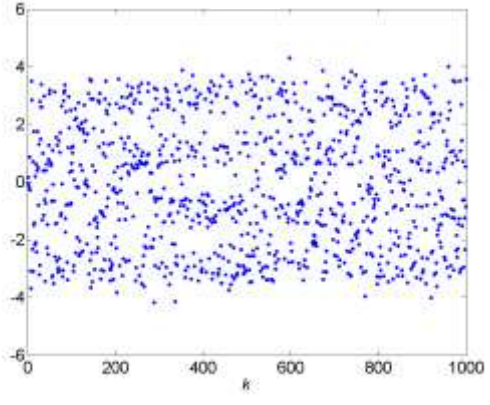




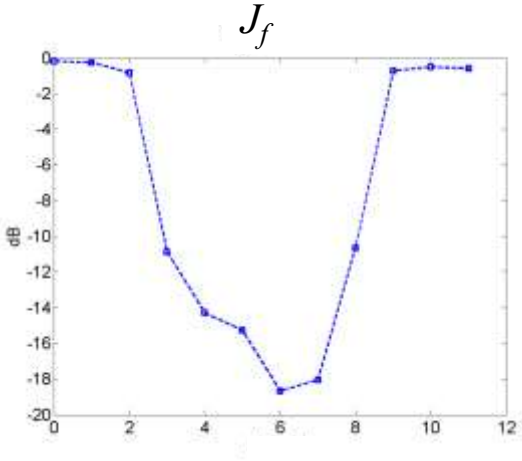
BPSK $\hat{s}[k]$

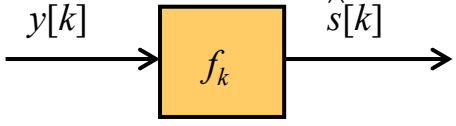


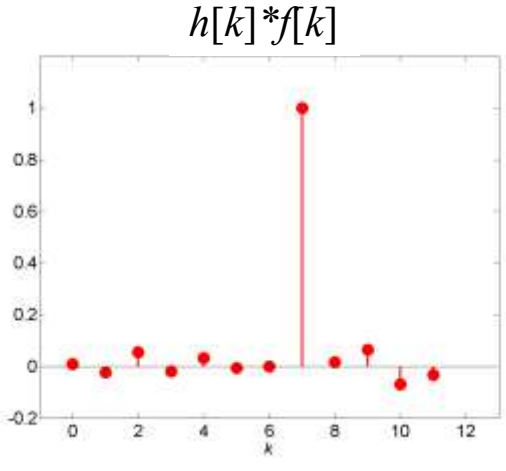
4_PAM $\hat{s}[k]$

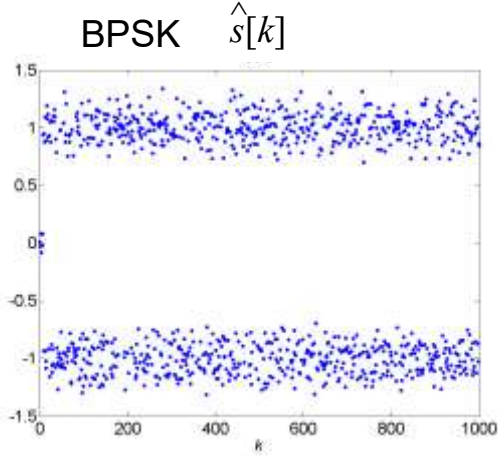


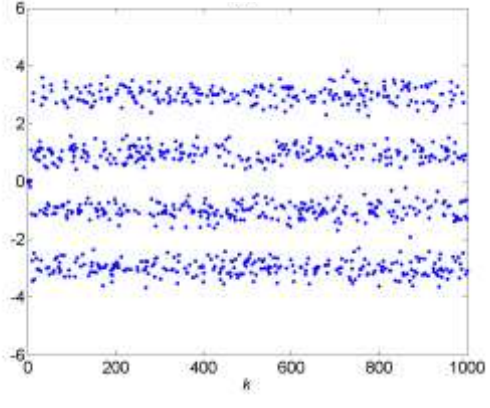
Example: LS equalizer, order 5



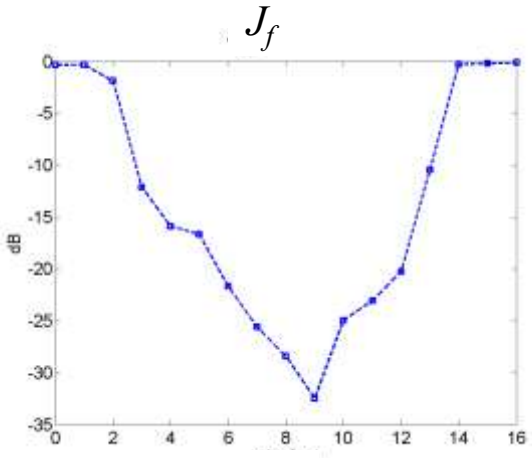




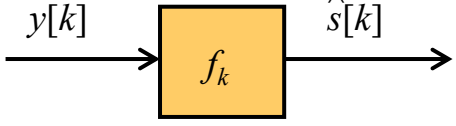


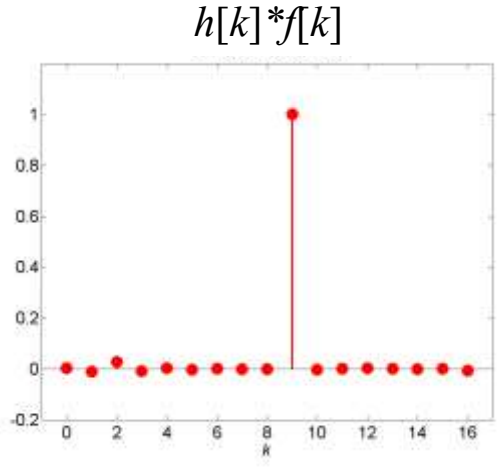


Example: LS equalizer, order 10

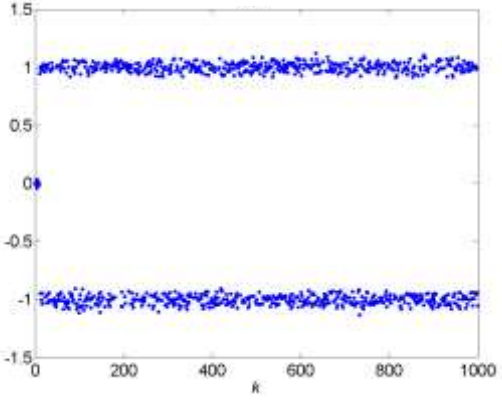


optimal delay
$n_d = 9$

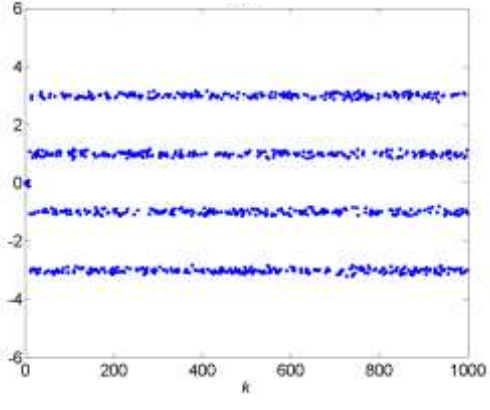




BPSK $\hat{s}[k]$



4_PAM $\hat{s}[k]$



Least squares equalizer

Learning objectives:

- Develop LS equalizers to remove the effects of intersymbol interference

Background on linear minimum mean squared error (LMMSE) estimation

Learning objectives:

- Derive the LMMSE estimator

Background on LMMSE estimator

- Suppose we want to estimate $\mathbf{x}$ from an observation $\mathbf{y}$
  - Unknown vector $\mathbf{x}$ of size M x 1 with zero mean and covariance $\mathbf{C}_{xx}$
  - Observation vector $\mathbf{y}$ with zero mean and covariance $\mathbf{C}_{yy}$
  - $\mathbf{x}$ and $\mathbf{y}$ are jointly correlated with covariance matrix $\mathbf{C}_{yx}$
- The objective of the LMMSE estimator is to determine a linear transformation such that

$$\mathbf{G}_{\text{MMSE}} = \arg\min_{\mathbf{G}} \mathbb{E}\left[\|\mathbf{x} - \mathbf{G}^*\mathbf{y}\|^2\right]$$

- Equivalently

$$\mathbf{G}_{\text{MMSE}} = \arg\min_{\mathbf{G}} \mathbb{E}\left[\sum_{m=1}^{M} |\mathbf{x}_m - \mathbf{g}_m^*\mathbf{y}|^2\right]$$

$\mathbf{x}_m = [\mathbf{x}]_m \qquad \mathbf{g}_m = [\mathbf{G}]_{:,m}$

LMMSE estimator

- Solving for one column of $\mathbf{G}_{\text{MMSE}}$

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}\mathbf{g}_k^{\mathrm{c}}}\mathbb{E}\left[\sum_{m=1}^{M}|\mathbf{x}_m-\mathbf{g}_m^*\mathbf{y}|^2\right] &= \mathbb{E}\left[\frac{\mathrm{d}}{\mathrm{d}\mathbf{g}_k^{\mathrm{c}}}\sum_{m=1}^{M}|\mathbf{x}_m-\mathbf{g}_m^*\mathbf{y}|^2\right]\\ &= \mathbb{E}\left[\frac{\mathrm{d}}{\mathrm{d}\mathbf{g}_k^{\mathrm{c}}}|\mathbf{x}_k-\mathbf{g}_k^*\mathbf{y}|^2\right]\\ &= \mathbb{E}\left[\mathbf{y}\left(\mathbf{y}^*\mathbf{g}_k-\mathbf{x}_k^*\right)\right].\end{aligned}$$

- MMSE orthogonality equation
  - Taking the expectation and setting the result to zero

$$\mathbf{C}_{\mathbf{yy}}\mathbf{g}_k = [\mathbf{C}_{\mathbf{yx}}]_{:,k}.$$

  - Solution is

$$\mathbf{g}_{k,\text{MMSE}} = \mathbf{C}_{\mathbf{yy}}^{-1}[\mathbf{C}_{\mathbf{yx}}]_{:,k}$$

LMMSE estimator

- Reassembling the column of $\mathbf{G}$ and combining the results together

$$\mathbf{G}_{\mathrm{MMSE}} = \mathbf{C}_{\mathbf{yy}}^{-1}\mathbf{C}_{\mathbf{yx}}$$

- The MMSE estimate of $\mathbf{x}$ is

$$\begin{aligned}\mathbf{x}_{\mathrm{MMSE}} &= \mathbf{G}_{\mathrm{MMSE}}^{*}\mathbf{y} \\ &= \mathbf{C}_{\mathbf{yx}}^{*}\mathbf{C}_{\mathbf{yy}}^{-1}\mathbf{y}.\end{aligned}$$

Exploits the joint correlation to extract information about $\mathbf{x}$ from $\mathbf{y}$

Decorrelates $\mathbf{y}$

Background on the linear minimum mean squared error (LMMSE) estimation

Learning objectives:

- Derive the LMMSE estimator

LMMSE equalizer

Learning objectives:

- Derive and compute the LMMSE equalizer

Reformulating the equalization problem

- Consider the received signal after the equalizer

$$\hat{s}[n-n_{\mathrm{d}}] = \sum_{\ell=0}^{L_{\mathrm{f}}} f_{n_{\mathrm{d}}}[\ell] y[n-\ell] \quad \Longrightarrow \quad \hat{s}[n-n_{\mathrm{d}}] = \mathbf{f}_{n_{\mathrm{d}}}^{\mathrm{T}} \mathbf{y}[n]$$

where

$$\mathbf{y}^{\mathrm{T}}[n] = [y[n], y[n-1], \ldots, y[n-L_{\mathrm{f}}]^{\mathrm{T}}$$

$$\mathbf{y}[n] = \mathbf{H}^{\mathrm{T}} \mathbf{s}[n] + \mathbf{v}[n]$$

$$\mathbf{s}^{\mathrm{T}}[n] = [s[n], s[n-1], \ldots, s[n-L]]^{\mathrm{T}}$$

- The LMMSE equalizer minimizes the error

$$\mathbb{E}\left[\left|s[n-n_{\mathrm{d}}] - \mathbf{f}_{n_{\mathrm{d}}}^{\mathrm{T}} \mathbf{y}[n]\right|^2\right]$$

Solving for the LMMSE equalizer

- Assuming that
  - $s[n]$ is IID with zero mean and unit variance,
  - $v[n]$ is IID with variance σ_v^2
  - $s[n]$ and $v[n]$ are independent
- Then

$$\mathbf{C}_{\mathbf{yy}} = \mathbb{E}\left[\mathbf{y}[n]\mathbf{y}^*[n]\right] = \mathbf{H}^{\mathrm{T}}\mathbf{H}^{\mathrm{c}} + \sigma_v^2\mathbf{I} \qquad \mathbf{C}_{\mathbf{y}s} = \mathbb{E}\left[\mathbf{y}[n]s^*[n-n_{\mathrm{d}}]\right] = \mathbf{H}^{\mathrm{T}}\mathbf{e}_{n_{\mathrm{d}}}.$$

- The LMMSE equalizer is computed obtaining the MMSE estimate that minimizes

$$\mathbb{E}\left[\left|s[n-n_{\mathrm{d}}] - \mathbf{f}_{n_{\mathrm{d}}}^{\mathrm{T}}\mathbf{y}[n]\right|^2\right]$$

$$\mathbf{f}_{n_{\mathrm{d}},\mathrm{MMSE}} = \mathbf{C}_{\mathbf{yy}}^{-\mathrm{c}}\mathbf{C}_{\mathbf{y}s}^{\mathrm{c}} = \left(\mathbf{H}^*\mathbf{H} + \sigma_v^2\mathbf{I}\right)^{-1}\mathbf{H}^*\mathbf{e}_{n_{\mathrm{d}}}$$

Linear equalization discussion

- Connections between LMMSE and LS solutions

$$\mathbf{f}_{n_\mathrm{d},\mathrm{MMSE}} = \left(\mathbf{H}^*\mathbf{H} + \sigma_v^2\mathbf{I}\right)^{-1}\mathbf{H}^*\mathbf{e}_{n_\mathrm{d}}$$

noise variance becomes large

$$\frac{1}{\sigma_v^2}\mathbf{H}^*\mathbf{e}_{n_\mathrm{d}}$$

matched filter

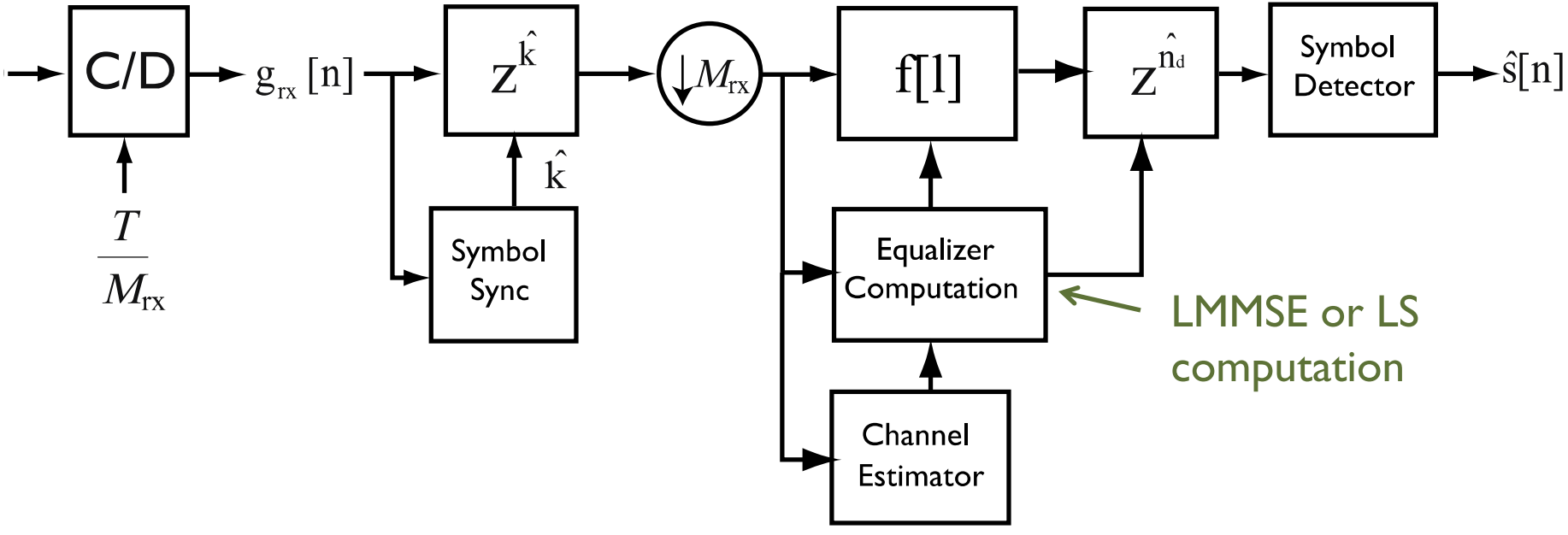
noise variance goes to zero

$$\mathbf{f}_{\mathrm{LS},n_\mathrm{d}} = (\mathbf{H}^*\mathbf{H})^{-1}\mathbf{H}^*\mathbf{e}_{n_\mathrm{d}}$$

LS solution

- Complexity of the equalizer depends on the choice of $\mathrm{L_f}$
  - Larger values give better performance but higher complexity
  - Generally take $\mathrm{L_f}$=4L is a reasonable compromise

Receiver with channel estimation and linear equalization



LMMSE equalizer

Learning objectives:

- Derive and compute the LMMSE equalizer

Background on the discrete Fourier transform

Learning objectives:

- Calculate the DFT and IDFT of a sequence
- Calculate the circular convolution of two sequences.

Frequency domain linear equalization

- Received signal with intersymbol interference without noise

$$y[n] = \sum_{\ell=0}^{L} h[\ell] s[n-\ell]$$

- In the DTFT domain $\mathsf{y}\left(e^{\mathrm{j}2\pi f}\right) = \mathsf{h}\left(e^{\mathrm{j}2\pi f}\right)\mathsf{s}\left(e^{\mathrm{j}2\pi f}\right)$

- Ideal zero-forcing equalizer is $\mathsf{f}\left(e^{\mathrm{j}2\pi f}\right) = \dfrac{1}{\mathsf{h}\left(e^{\mathrm{j}2\pi f}\right)}$

  Must be computed for all continuous frequency values

Equalization in the frequency domain allows perfect inverse of the channel to be computed

Need to think in a different frequency domain

Discrete Fourier Transform (DFT)

- The DFT is a basis expansion for finite-length signals

$$\text{Analysis} \quad \mathrm{x}[k] = \sum_{n=0}^{N-1} x[n] e^{-\mathrm{j}\frac{2\pi}{N}kn} \quad k = 0, 1, ..., N-1$$

$$\text{Synthesis} \quad x[n] = \frac{1}{N} \sum_{k=0}^{N-1} \mathrm{x}[k] e^{\mathrm{j}\frac{2\pi}{N}kn} \quad n = 0, 1, ..., N-1$$

- The DFT can be computed efficiently with the Fast Fourier transform for N a power of 2 and certain other special cases

circular shift

circular convolution

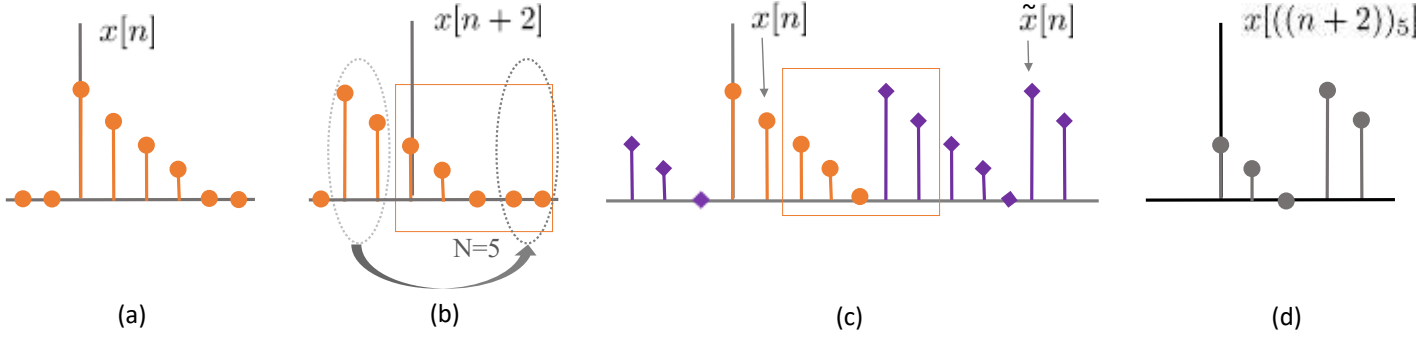
Finite-Length Sequence (Length N)	N-point DFT (Length N)
$x[n]$	$\mathsf{x}[k]$
$x_1[n], x_2[n]$	$\mathsf{x}_1[k], \mathsf{x}_2[k]$
$ax_1[n] + bx_2[n]$	$a\mathsf{x}_1[k] + b\mathsf{x}_2[k]$
$\mathsf{x}[n]$	$Nx[((-k))_N]$
$x[((n-m))_N]$	$W_N^{km}\mathsf{x}[k]$
$W_N^{-\ell n}x[n]$	$\mathsf{x}[((k-\ell))_N]$
$\sum_{m=0}^{N-1} x_1[m]x_2[((n-m))_N]$	$\mathsf{x}_1[k]\mathsf{x}_2[k]$
$x_1[n]x_2[n]$	$\frac{1}{N}\sum_{l=0}^{N-1} \mathsf{x}_1[\ell]\mathsf{x}_2[((k-\ell))_N]$
$x^*[n]$	$\mathsf{x}^*[((-k))_N]$
$x^*[((-n))_N]$	$\mathsf{x}^*[k]$
$\mathrm{Re}\{x[n]\}$	$X_{\mathrm{ep}}[k] = \frac{1}{2}\{\mathsf{x}[((k))_N] + \mathsf{x}^*[((-k))_N]\}$
$\mathrm{jIm}\{x[n]\}$	$X_{\mathrm{op}}[k] = \frac{1}{2}\{\mathsf{x}[((k))_N] - \mathsf{x}^*[((-k))_N]\}$
$x_{\mathrm{ep}}[k] = \frac{1}{2}\{x[n] + x^*[((-n))_N]\}$	$\mathrm{Re}\{\mathsf{x}[k]\}$
$x_{\mathrm{op}}[k] = \frac{1}{2}\{x[n] - x^*[((-n))_N]\}$	$\mathrm{jIm}\{X[k]\}$
† $x[n]$ real	
Symmetry Properties	$\begin{cases} \mathsf{x}[k] = \mathsf{x}^*[((-k))_N] \\ \mathrm{Re}\{\mathsf{x}[k]\} = \mathrm{Re}\{\mathsf{x}[((-k))_N]\} \\ \mathrm{Im}\{\mathsf{x}[k]\} = -\mathrm{Im}\{\mathsf{x}[((-k))_N]\} \\ \lvert\mathsf{x}[k]\rvert = \lvert\mathsf{x}[((-k))_N]\rvert \\ \measuredangle\mathsf{x}[k] = -\measuredangle\mathsf{x}[((-k))_N] \end{cases}$
$x_{\mathrm{ep}}[k] = \frac{1}{2}\{x[n] + x[((-n))_N]\}$	$\mathrm{Re}\{\mathsf{x}[k]\}$
$x_{\mathrm{op}}[k] = \frac{1}{2}\{x[n] - x[((-n))_N]\}$	$\mathrm{jIm}\{\mathsf{x}[k]\}$

Circular shift

- Because of finite length property, shifts are circular

$$x[((n-m))_N] \longleftrightarrow W_N^{km}\mathrm{x}[k] \qquad W_N = \exp(-\mathrm{j}2\pi/N)$$

- Example of circular shift



Circular convolution

- Circular convolution

$$x_1[n] \circledast x_2[n] = \sum_{m=0}^{N-1} x_1[m]\, x_2[(n-m) \bmod N]$$

Circular wrapping

- In the frequency domain

$$x_1[n] \circledast x_2[n] \longleftrightarrow X_1[k]X_2[k]$$

- To equalize in the frequency domain, it will be necessary to replicate the effects of the circular convolution

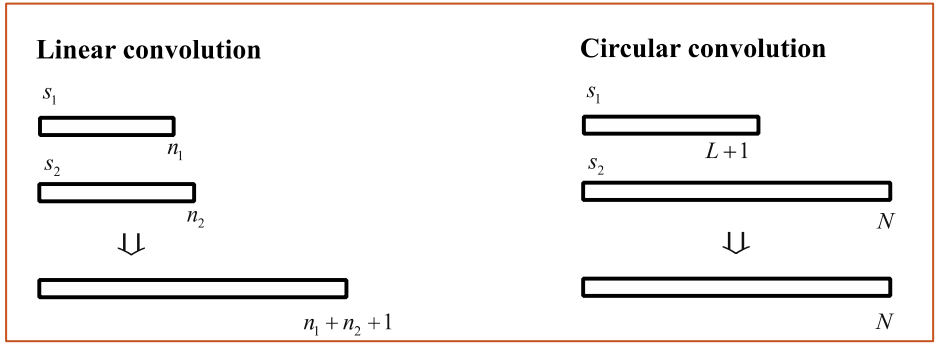
Circular convolution with a zero-padded sequence

- To illustrate, let length of $h[n]$ be $L+1$, length of $s[n]$ be $N > L$
- Zero-pad $h[n]$ with $N-1-L$ zeros

$$y[n] = \sum_{\ell=0}^{N-1} h[\ell] s\left[((n-\ell))_N\right]$$

$$= \sum_{\ell=0}^{L} h[\ell] s\left[((n-\ell))_N\right]$$

$$= \begin{cases} \sum_{\ell=0}^{n} h[\ell] s[n-\ell] + \sum_{\ell=N+1}^{L} h[\ell] s[n+n-\ell] & 0 \leqslant n < L \\ \sum_{\ell=0}^{L} h[\ell] s[n-\ell] & n \geqslant L \end{cases}$$



Circular wrapping effect

Looks linear

Background on the discrete Fourier transform

Learning objectives:

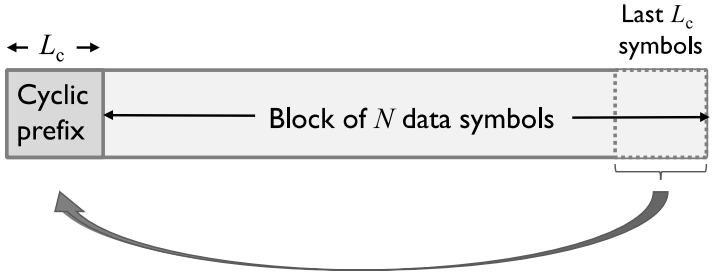
- Calculate the DFT and IDFT of a sequence
- Calculate the circular convolution of two sequences

Single carrier frequency domain equalization

Learning objectives:

- Use cyclic prefix to convert a linear convolution to a circular convolution and equalize in frequency domain

Cyclic prefix



- Organize data into blocks of N symbols
- Assemble a block of N + L_c symbols where
  - Prefixed part $w[n] = s[n + N - L_c] \quad n = 0, 1, \ldots, L_c - 1$
  - Data part $w[n] = s[n - L_c] \quad n = L_c, L_c + 1, \ldots, L_c + N - 1$
- Neglect the first Lc terms of the convolution

$$\bar{y}[n] = y[n + L_c] \quad n = 0, 1, \ldots, N - 1$$
$$= \sum_{\ell=0}^{L} h[\ell] w[n + L_c - \ell].$$

Exposing the circular convolution 1/3

$$\bar{y}[0] = \sum_{\ell=0}^{L} h[\ell] w[0 + L_{\mathrm{c}} - \ell] \tag{119}$$

$$= h[0]w[L_{\mathrm{c}}] + h[1]w[L_{\mathrm{c}} - 1] + \cdots + h[\ell]w[L_{\mathrm{c}} - \ell] \tag{120}$$

$$= h[0]s[0] + h[1]s[N - 1] + \cdots + h[\ell]s[N - L - 1] \tag{121}$$

$$= \sum_{\ell=0}^{L} h[\ell] s[((0 - \ell))_N] \tag{122}$$

Exposing the circular convolution 2/3

$$\bar{y}[L-1] = \sum_{\ell=0}^{L} h[\ell] w[L-1+L_{\mathrm{c}}-\ell] \tag{123}$$

$$= h[0]w[L-1+L_{\mathrm{c}}] + h[1]w[L-1+L_{\mathrm{c}}-1] + \cdots + h[\ell]w[L-1+L_{\mathrm{c}}-\ell] \tag{124}$$

$$= h[0]s[L-1] + h[1]s[L-2] + \cdots + h[L-1]s[0] + h[\ell]s[N-1] \tag{125}$$

$$= \sum_{\ell=0}^{L} h[\ell] s[((L-1-\ell))_N] \tag{126}$$

Exposing the circular convolution 3/3

$$\bar{y}[L] = \sum_{\ell=0}^{L} h[\ell] w[L + L_c - \ell] \tag{127}$$

$$= h[0]w[L + L_c] + h[1]w[L + L_c - 1] + \cdots + h[\ell]w[L + L_c - \ell] \tag{128}$$

$$= h[0]s[\ell] + h[1]s[\ell] + \cdots + h[\ell]s[0] \tag{129}$$

$$= \sum_{\ell=0}^{L} h[\ell] s[((L - \ell))_N] \tag{130}$$

$$= \sum_{\ell=0}^{L} h[\ell] s[L - \ell]. \tag{131}$$

Connection to frequency domain equalization

- Conclude that the received signal is equivalently

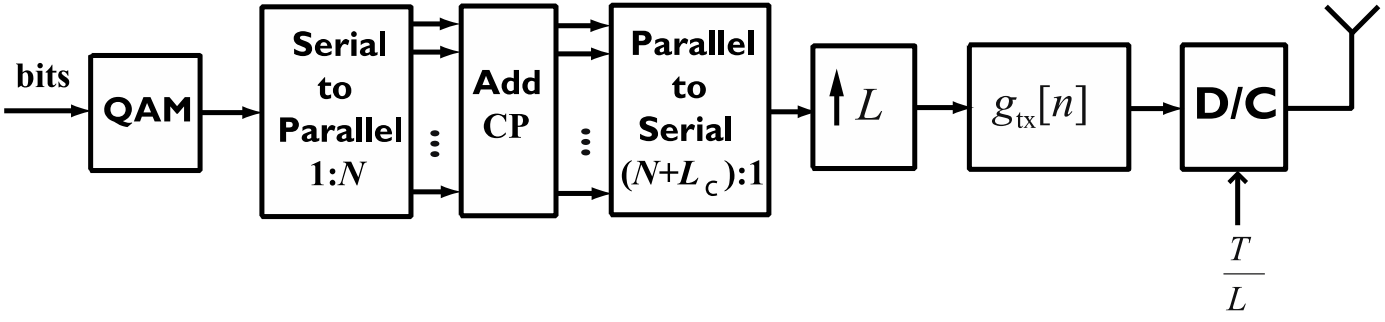
$$\bar{y}[n] = \sum_{\ell=0}^{L} h[\ell] s[((n-\ell))_N]$$

- Using the DFT equalization can proceed as

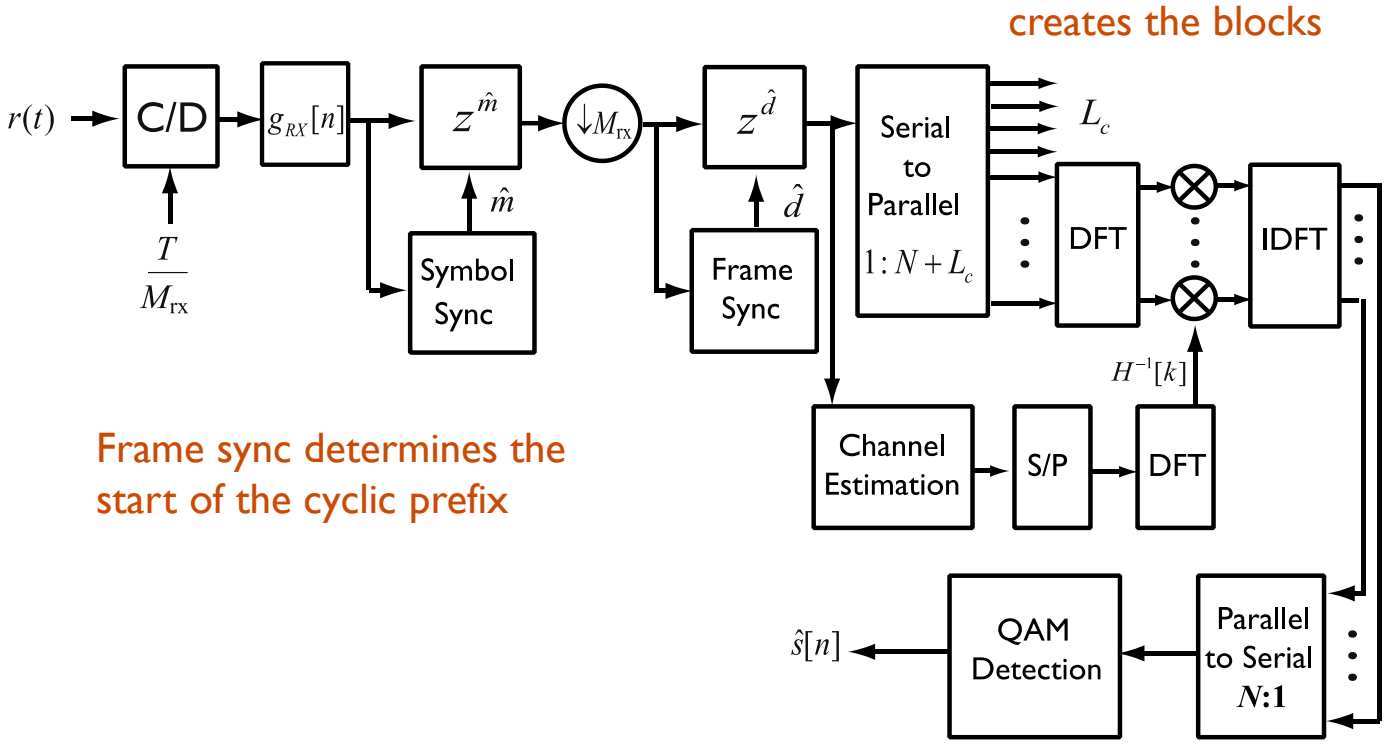
$$\begin{aligned}\hat{s}[n] &= \mathcal{F}_N^{-1}\left(\frac{\mathsf{y}[k]}{\mathsf{h}[k]}\right) \\ &= \mathcal{F}_N^{-1}\left(\frac{\mathcal{F}_N(y[n])}{\mathcal{F}_N(h[n])}\right)\end{aligned}$$

- Can use frequency domain equalization!

SC-FDE transmitter



SC-FDE receiver



Single carrier frequency domain equalization

Learning objectives:

- Use cyclic prefix to convert a linear convolution to a circular convolution and equalize in frequency domain

6G Technology Video

https://www.ni.com/en/solutions/5g-6g.html?srsltid=AfmBOooHu-zOxAoA5kNKZ8INvhcVzw4p52iA2a4ZQcF-WTjCBxs5xLu5